

Nemotron-CC: Transforming Common Crawl into a Refined Long-Horizon Pretraining Dataset

D. Su, K. Kong, Y. Lin, J. Jennings, B. Norick, M. Kliegl, M. Patwary, M. Shoeybi, and B. Catanzaro, 'Nemotron-CC: Transforming Common Crawl into a Refined Long-Horizon Pretraining Dataset', May 30, 2025, arXiv: arXiv:2412.02595. doi: 10.48550/arXiv.2412.02595.

NEMOTRON-CC ABSTRACT

Metrics: MMLU = Massive Multitask Language Understanding; Score = percentage of correctly answered multiple-choice questions (0-100 scale) - **Problem:** Aggressive filtering in FineWeb-Edu, DCLM removes ~90% of data → unsuitable for long-horizon training (15T tokens) - **Solution:** Classifier ensembling + synthetic rephrasing + reduced heuristic filtering - **Results:** - Short horizon (1T tokens, 8B): - High-quality subset: +5.6 pp MMLU vs. DCLM - Long horizon (15T tokens, 8B, 7.2T from Nemotron-CC): - vs. Llama 3.1 8B: +5.0 pp MMLU (70.3 vs. 65.3) - +3.1 pp ARC-Challenge - +0.5 pp average (10 tasks) - Data efficiency (1T tokens): - Full dataset: parity MMLU with DCLM, but 4× more unique real tokens

- **Composition (6.3T total):**

- Deduplicated original tokens: 4.4T
- Synthetic tokens: 1.9T

INTRODUCTION

- Context: Internet crawl = largest source of unique tokens for LLMs
 - Two purposes: high-quality content + diversity

- Status quo (FineWeb-Edu, DCLM):
 - Use model-based filters for high-quality content
 - Boost benchmark accuracies
 - Cost: remove ~90% of data
- Problem - Long-horizon training bottleneck:
 - Recent models train on extended token horizons
 - Llama 3.1 (8B–405B): 15T tokens
 - Gemma 2 27B: 13T tokens
 - Aggressive pruning creates data scarcity for extended training
 - DCLM & FineWeb-Edu contain ~80% near-duplicates (1T and 0.2T unique tokens respectively)
 - Training many epochs on same data = diminishing returns (Muennighoff et al., 2024); better to train on more unique tokens
- Proposed solution: Better accuracy-quantity trade-off via:
 - Classifier ensembling
 - Synthetic data generation
 - Reduced heuristic filter reliance
- Contributions:
 - 6.3T token long-horizon dataset: 4.4T deduplicated original + 1.9T synthetic
 - Released under Common Crawl Terms of Use (CC contributor framework)
 - NeMo Curator library (Apache 2.0): open-source data curation toolkit with reference implementation
 - Quality classifier models released
 - Effectiveness proof vs. DCLM/FineWeb-Edu:
 - 1.1T high-quality subset: +5.6 pp MMLU
 - Full dataset: parity MMLU vs. DCLM, but 4× more unique real tokens
 - 8B @ 15T tokens: +5.0 pp MMLU vs. Llama 3.1 8B (70.3 vs. 65.3)
 - Ablations show:
 - Ensembling selects larger, more diverse high-quality tokens
 - Rephrasing reduces noise + produces fresh variants
 - Disabling heuristic filters boosts high-quality token yield
- Guiding principle: Shift from static heuristic pipeline → learned flywheel
 - Improved data → better LLMs → better synthetic data & quality classification (virtuous cycle)

METHODS

- Overview: Three-component approach to build English Common Crawl pretraining dataset
 - HTML-to-text extraction & heuristic filters (token yield)
 - Model-based quality labeling pipeline
 - Synthetic data generation
- Full pipeline schematic: Figure 3 (Appendix A)

2.1 HTML-to-text Extractor & Filter

- Foundation: Text extraction from HTML is major source of pretraining data
- Reference standard: FineWeb-Edu classifier (model-based quality classifier, proven effective)
- Goal: Balance data quality against token yield, avoid removing good tokens via over-filtering

HTML-to-text Extraction

- Tested: Justext (Pomikálek, 2011) vs. Trafilatura (Barbaresi, 2021)
- Qualitative assessment: both same level
- Quantitative (13 Common Crawl snapshots, Table 1):
 - Justext yields more tokens overall
 - Justext high-quality tokens: +28.6% (by FineWeb-Edu standard: scores 3–5)
 - Strategy: Maximize absolute HQ token count, not percentage
 - Justext: 127B HQ tokens (7.0%)
 - Trafilatura: 104B HQ tokens (7.5%)
- Language filtering: pylld2 + FastText language identifier (threshold 0.3)
- Deduplication:
 - Global fuzzy deduplication
 - Exact substring deduplication across eighths of snapshots
 - Tools: NeMo Curator library + deduplicate-text-datasets library

Filtering

- Conventional heuristic pipeline (from Parmar et al., 2024):
 - Raffel et al. (2020) + Rae et al. (2021) heuristics
 - KenLM perplexity filter (trained on Wikipedia + books)
- Problem: Heuristic filters remove 18.1% of high-quality tokens (FineWeb-Edu standard)

- Proposal: Apply heuristic filters selectively
 - Do NOT apply to high-quality tokens (identified by model-based classifiers)
 - Apply only to low-quality splits
- Validation: Ablation studies (Section 3.3)

Model-based Quality Labeling

- Problem: Existing classifiers (DCLM, FineWeb-Edu) have limited recall (~10% high-quality tokens)
 - Bottleneck for long-horizon training
 - Quality labels not necessarily aligned with LLM downstream task performance
- Solution: Ensemble-based quality labeling pipeline
 - Build three classifiers with different quality preferences
 - Ensemble scores across all documents
 - Split corpus into quality buckets
 - Regroup into 5 quality levels based on downstream task performance

Quality Classifier Training

- Two annotation versions: prompt Mistral 8x22B-instruct + Nemotron-340B-instruct
 - Score documents from FineWeb on educational value (0–5 scale)
- Fine-tune linear regression models on Snowflake-arctic-embed-m embeddings (text-to-vector conversion model)
 - Frozen embedding + encoder layers (weights unchanged during training, only linear layer updated)
 - 20 epochs, learning rate 0.0003
 - Checkpoint selected by highest F1 on validation set
- Third classifier: DCLM classifier (fastText-based, Li et al., 2024)
 - Trained on instruction-formatted data + ELI5 subreddit high-scoring posts (Explain-
Like-I’m-5: community-curated, high-quality explanations)
 - Stronger performance identifying high-quality tokens vs. FineWeb-Edu
 - Offers different perspective for increased recall

Quality Scoring and Bucketing

- Each classifier predicts quality scores for all documents

- Round scores to integers 0–19 (20 equal-sized buckets, ~5% of documents per bucket)
 - Bucket 19 = top 5% highest quality
- Final quality score: ensemble via maximum operation across three classifiers
 - Document distribution across buckets skewed by ensemble operation

Quality Labeling

- Map fine-grained scores (0–19) to 5 downstream quality categories
- Method: continuous pretraining annealing
 - Setup: 50B token training on 70% pretrained 8B model
 - Metric: average performance across 9 tasks per bucket
 - Weighted mix: 66% standard baseline data + 34% evaluated bucket
 - (Tests each bucket's contribution; higher gain = higher quality category)
- Group 20 buckets into 5 quality categories based on performance

Synthetic Data Generation

- Observation: Low-quality docs contain noise/errors; high-quality have good writing/formatting
- Strategy: Different approaches for low vs. high quality

Low-Quality Data

- Goal: Reduce noise/errors while preserving useful information (cut training compute)
- Method: Rephrasing using medium-sized LM (Maini et al., 2024)
 - NOT creating new knowledge (unlike textbook/story generation approaches)
 - Transform text to Wikipedia style (Prompt 5, Appendix H)
 - Reduces errors, redundancies, improves formatting
- Result: Enhanced parallel corpus, lower perplexity, better downstream performance

High-Quality Data

- Goal: Generate unique tokens, condense essential knowledge
 - Rationale: Repeated tokens show diminishing returns after 4 epochs (Muennighoff et al., 2024)

- Four synthesis prompts:
 - Diverse Question-Answer pairs: ask varied question types (yes/no, open-ended, multiple-choice) with correct answers
 - Distill: rewrite as concise, clear passage
 - Extract knowledge: rewrite only informative content, discard filler
 - Knowledge list: structure key information as organized list
- Requirements: Clear, concise responses preserving factual info + concrete details (numbers, etc.)

Processing

- Chunking: Split documents into segments (complete lines, token limit enforced)
 - Prevents over-simplified model outputs on long inputs
 - Discard over-length lines exceeding token limit
- Post-processing:
 - Remove incomplete results, markdown formatting (e.g., **bold**)
 - Strip prefixes (“Here is a paraphrased version:”, “Paraphrased Text:”)
 - Remove enclosing quotation marks
 - Filter under-length outputs (< 50 tokens)
 - Wikipedia results: concatenate passages from same original document
 - Diverse QA results: shuffle pairs, append to segment end

Scale & Implementation

- Model: Mistral NeMo 12B (instruct, FP8 inference)
 - top-p: 0.9, temperature: 0.5
- Synthesis (Table 3):
 - Low-quality (403.0B raw) → Wikipedia → 336.3B synthetic
 - High-quality (451.3B raw):
 - Wikipedia: 372.9B
 - Diverse QA Pairs: 499.5B
 - Distill: 157.6B
 - Extract Knowledge: 303.6B
 - Knowledge List: 203.2B
 - Total synthetic: 1.8T tokens
 - Medium-quality: not synthesized (resource constraints)

- Tools: TensorRT-LLM + NeMo-Skills

Putting It All Together

- Combine all techniques above on 99 Common Crawl snapshots (CC-MAIN-2013-20 through CC-MAIN-2024-30)
- **Nemotron-CC dataset: 6.3T tokens**
 - 4.4T globally deduplicated original tokens
 - 1.9T synthetically derived tokens
- Comparison to baselines:
 - ~4× more unique tokens vs. FineWeb-Edu-2 and DCLM
 - Reason: DCLM/FineWeb used sharded approximate deduplication only, contain ~80% fuzzy duplicates
- High-quality subset (for fairer short-horizon comparison):
 - Nemotron-CC-HQ: 1.1T tokens
 - Composition: highest-scoring real data + diverse QA pair synthetic variants only

Dataset sizes in trillions of tokens. “Unique” shows the estimated number of tokens after global fuzzy deduplication of the real tokens.

Dataset sizes in trillions of tokens. “Unique” shows the estimated number of tokens after global fuzzy deduplication of the real tokens.